Problem Set 7: SVD and PCA

AI 501: Mathematics for Artificial Intelligence

Fall 2026

Show your working for calculation questions. Unless stated otherwise, use the sample mean to center data and divide by n when computing a covariance matrix, as in the PCA lecture. In the design questions, explain your choices and the trade-offs they create.

Problem 1. Consider the diagonal matrix

$$A = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

(a) Give matrices U, Σ, and V such that $A = U\Sigma V^T$.

(b) What are the singular values of A? Which input direction is stretched more by the transformation $x \mapsto Ax$?

Solution:

(a) Since A is diagonal with nonnegative entries, one valid decomposition is $U = I_2$, $\Sigma = A$, and $V = I_2$.

(b) The singular values are 3 and 1. The first coordinate direction, $(1, 0)^T$, is stretched by 3, while $(0, 1)^T$ is stretched by 1.

Problem 2. Let

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}.$$

(a) Compute $A^T A$ and find its eigenvalues and an orthonormal set of eigenvectors.

(b) Use your answer to identify the singular values of A.

(c) State the rank of A. How many nonzero singular values does it have?

Solution:

(a) $A^T A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. Its eigenvalues are 2 and 0, with unit eigenvectors $v_1 = (1, 1)^T/\sqrt{2}$ and $v_2 = (1, -1)^T/\sqrt{2}$.

(b) The singular values are the square roots of these eigenvalues: $\sqrt{2}$ and 0.

(c) The two columns of A are equal and nonzero, so $\operatorname{rank}(A) = 1$. It has one nonzero singular value.

Problem 3. Let

$$A = \begin{pmatrix} 3 & 0 \\ 4 & 0 \end{pmatrix}.$$

One SVD of this matrix is

$$A = U\Sigma V^T, \qquad U = \begin{pmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 5 & 0 \\ 0 & 0 \end{pmatrix}, \quad V = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

(a) Verify the factorization by multiplying the three matrices.

(b) For a unit vector $x = (x_1, x_2)^T$, what is $\|Ax\|$? Which input direction gives the largest output length?

(c) Describe geometrically what this matrix does to the plane.

Solution:

(a) Multiplication gives $U\Sigma V^T = \begin{pmatrix} 3 & 0 \\ 4 & 0 \end{pmatrix} = A$.

(b) $Ax = (3x_1, 4x_1)^T$, so $\|Ax\| = 5|x_1|$. For a unit input, the largest output length is 5, attained at $x = (1, 0)^T$ or its negative.

(c) The map sends every point to the line spanned by $(3, 4)^T$. It scales the first coordinate direction by 5 and collapses the perpendicular direction to zero.

Problem 4. Suppose a data matrix has singular values 10, 2, and 0.5.

(a) What is the rank of the matrix?

(b) If we keep only the largest singular value, what rank approximation do we obtain, and what is the Frobenius norm of the discarded part?

(c) Compute the fraction of the squared Frobenius norm retained by this rank-one approximation.

Solution:

(a) All three singular values are nonzero, so the rank is 3.

(b) Keeping only 10 gives a rank-one approximation. The discarded Frobenius norm is $\sqrt{2^2 + 0.5^2} = \sqrt{4.25} = \sqrt{17}/2$.

(c) The squared Frobenius norm retained is $10^2/(10^2 + 2^2 + 0.5^2) = 400/417 \approx 0.959$, or about 95.9%.

Problem 5. Four observations, each with two features, are

$$x_1 = (2, 1)^T, \quad x_2 = (0, -1)^T, \quad x_3 = (1, 2)^T, \quad x_4 = (-1, 0)^T.$$

(a) Find the sample mean and write the centered observations as rows of a matrix X_c.

(b) Compute the covariance matrix $C = \frac{1}{4} X_c^T X_c$.

(c) Find the eigenvalues and corresponding unit eigenvectors of C. Which direction is the first principal component?

Solution:

(a) The mean is $(1/2, 1/2)^T$. With rows as centered observations,

$$X_c = \begin{pmatrix} 3/2 & 1/2 \\ -1/2 & -3/2 \\ 1/2 & 3/2 \\ -3/2 & -1/2 \end{pmatrix}.$$

(b) $C = \frac{1}{4} X_c^T X_c = \begin{pmatrix} 5/4 & 3/4 \\ 3/4 & 5/4 \end{pmatrix}$.

(c) The eigenvalues are 2 and 1/2. Corresponding unit eigenvectors are $(1,1)^T/\sqrt{2}$ and $(1,-1)^T/\sqrt{2}$. The first principal component is $(1,1)^T/\sqrt{2}$.

Problem 6. Use the centered observations from Problem 5.

(a) Project each observation onto the first principal component to obtain its one-dimensional PCA score.

(b) Reconstruct each observation using only this score and the first principal component. Give the reconstructed points in the original two-feature coordinates.

(c) In words, describe what information this one-dimensional representation keeps and what it loses.

Solution: The first principal component is $v_1 = (1,1)^T/\sqrt{2}$. The scores, in the given observation order, are $\sqrt{2}, -\sqrt{2}, \sqrt{2}, -\sqrt{2}$. Reconstructing each centered point as $z_i v_1$ gives

$$\begin{pmatrix} 1 & 1 \\ -1 & -1 \\ 1 & 1 \\ -1 & -1 \end{pmatrix}.$$

These are centered-coordinate reconstructions; add the mean $(1/2, 1/2)^T$ to each row to return to the original coordinates. The representation preserves variation along the diagonal direction and discards variation along $(1,-1)^T/\sqrt{2}$.

Problem 7. The following centered observations describe four objects using two measurements:

$$X_c = \begin{pmatrix} 2 & 0 \\ 0 & 1 \\ -2 & 0 \\ 0 & -1 \end{pmatrix}.$$

(a) Compute the covariance matrix $C = \frac{1}{4} X_c^T X_c$ and its eigenvalues.

(b) What fraction of the total variance is retained by keeping only the first principal component?

(c) A researcher wants a two-dimensional view for visualization. Should they keep one or two components? Explain the trade-off using your results.

Solution:

$$C = \frac{1}{4} X_c^T X_c = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix},$$

so the eigenvalues are 2 and 1/2. Keeping the first component retains $2/(2 + 1/2) = 0.8$, or 80%, of the variance. Keep one component for a one-dimensional compressed representation; keep two for the requested two-dimensional visualization, which preserves all variance but does not reduce dimension.

Problem 8. A centered data matrix has SVD $X_c = U\Sigma V^T$. Explain the relationship between the PCA directions and the columns of V by showing how

$$C = \frac{1}{n} X_c^T X_c$$

can be written using V and Σ. If the singular values are $\sigma_1 \geq \sigma_2 \geq \cdots$, which columns of V should be used for a k-dimensional PCA representation? Explain briefly.

Solution: Using $X_c = U\Sigma V^T$ and $U^TU = I$, we obtain $C = V(\Sigma^T\Sigma/n)V^T$. Thus the columns of V are covariance eigenvectors, with eigenvalues σ_i^2/n. Use the first k columns of V, which correspond to the largest singular values and therefore the largest variances.

Problem 9. An image is represented by a matrix $A \in \mathbb{R}^{m\times n}$ of pixel intensities. Its rank-k SVD approximation is $A_k = U_k\Sigma_k V_k^T$.

(a) How many real numbers are needed to store A directly? How many are needed to store the three factors in A_k, ignoring any small storage savings from matrix structure?

(b) For an image of size 1000×800, compare these counts when $k = 20$.

(c) Explain why increasing k generally improves image fidelity but reduces compression.

Solution:

(a) Direct storage uses mn numbers. A rank-k representation uses $mk + k + kn = k(m + n + 1)$ numbers.

(b) Direct storage uses 800,000 values. With $k = 20$, the factors use $20(1000 + 800 + 1) = 36{,}020$ values, about 4.5% of the original count.

(c) Adding singular components can only reduce or preserve the best rank-k approximation error, so it tends to preserve more image detail. But each added component requires $m + n + 1$ additional values, reducing the storage savings.

Problem 10. You are asked to build a compact representation of a dataset containing thousands of customer records, each described by hundreds of numeric measurements. The downstream model is too slow with all measurements.

(a) Describe a PCA-based preprocessing pipeline from raw measurements to reduced features. Include how you handle centering and how you choose the number of components.

(b) Two measurements use very different units (annual income in dollars and age in years). What should you check before applying PCA, and what choice would you make if scale differences are not meaningful?

(c) Give one measure of information retained and one downstream check you would use to decide whether the reduction is useful.

Solution:

(a) Split the data into training and evaluation sets. Estimate feature means on the training set, center the training data, and apply those same means to the evaluation data. Fit PCA on the centered training data, project both sets onto the first k components, and train the downstream model on the reduced training features. Choose k using a target fraction of variance retained or validation performance.

(b) Check whether the different units should affect the analysis. If scale differences are not meaningful, standardize each feature using its training-set mean and standard deviation before PCA, then apply those same values to evaluation data.

(c) Measure the fraction of variance retained. Compare downstream validation performance and computation time with the model using all original features.

Problem 11. A team wants to compress a collection of face photographs using a low-rank representation. Propose a practical SVD-based design.

(a) Explain how you would arrange the pixel data into a matrix and how the SVD would produce a compressed representation.

(b) What quantity would you inspect to choose the rank k? Give one reason not to choose k solely by maximizing visual quality.

(c) Describe how you would compare the compressed images with the originals, and name one limitation of low-rank compression for photographs.

Solution:

(a) Flatten every face image in the same pixel order and place each flattened image in a row of a data matrix. Compute its SVD and store the first k columns of U, the first k singular values, and the first k rows of V^T; their product reconstructs an approximate set of faces.

(b) Inspect the cumulative squared-singular-value fraction, $\sum_{i=1}^{k} \sigma_i^2 / \sum_i \sigma_i^2$, and choose k to meet a storage or quality target. Do not choose k solely for visual quality: a larger k may look better but require more storage, and visual fidelity may not reflect usefulness for face recognition.

(c) Compare reconstructions using relative Frobenius error and visual inspection. Low-rank compression may blur fine details or remove subtle features important for recognition.

Problem 12. A movie service has a user-by-movie table of ratings, but most entries are missing because users have not watched most movies. The team wants to predict some missing ratings using a low-rank model.

(a) Explain why directly applying an ordinary SVD to a table that replaces every missing rating by zero can give misleading results.

(b) Propose a reasonable workflow that uses low-rank structure while respecting which ratings were actually observed. State what the model should be fitted to and how its rank could be selected.

(c) Name a suitable validation strategy and one metric for judging rating predictions. Explain why the training error alone is not enough.

Solution:

(a) Replacing missing entries by zero treats "not observed" as a genuine zero rating. The many inserted zeros can distort the matrix structure and dominate the decomposition.

(b) Fit a low-rank factor model by minimizing prediction error only over the observed user–movie pairs, with regularization to limit overfitting. Select the rank and regularization strength using validation data.

(c) Hold out some observed ratings (or use a time-based split when predicting future ratings) and evaluate predictions with root mean squared error. Training error alone is insufficient because it can decrease as model complexity grows even while predictions on unseen ratings get worse.

Problem 13. (Optional coding; not examinable) Choose a small grayscale image and represent it as a matrix of pixel values. Using a programming language and numerical library of your choice:

(a) Compute its SVD and reconstruct it using the largest k singular values, for example $k = 5$, 20, and 50 (adjust these if needed for your image).

(b) Display the original and reconstructed images side by side.

(c) For each k, report the storage count $k(m+n+1)$ and the relative reconstruction error $\|A - A_k\|_F / \|A\|_F$. Briefly describe how the image changes as k increases.

Solution: Run the following code snippets in order in the same Python session. They use NumPy for the SVD, Pillow to load a grayscale image, and Matplotlib to display the results. Replace 'image.png' with the path to your image.

(a) Load the image as a matrix, compute its SVD, and reconstruct it using the selected singular values:

```
import numpy as np
from PIL import Image

A = np.asarray(Image.open("image.png").convert("L"), dtype=float)
m, n = A.shape
U, s, Vt = np.linalg.svd(A, full_matrices=False)
ks = [k for k in (5, 20, 50) if k <= len(s)]
approximations = []
for k in ks:
    Ak = (U[:, :k] * s[:k]) @ Vt[:k, :]
    approximations.append((k, Ak))
```

(b) Display the original image beside the reconstructions:

```
import matplotlib.pyplot as plt

fig, axes = plt.subplots(1, len(ks) + 1, figsize=(4 * (len(ks) + 1), 4))
axes[0].imshow(A, cmap="gray", vmin=0, vmax=255)
axes[0].set_title("Original")
axes[0].axis("off")

for ax, (k, Ak) in zip(axes[1:], approximations):
    ax.imshow(Ak, cmap="gray", vmin=0, vmax=255)
    ax.set_title(f"k={k}")
    ax.axis("off")

plt.tight_layout()
plt.show()
```

(c) Report the storage count and relative reconstruction error:

```
for k, Ak in approximations:
    error = np.linalg.norm(A - Ak, "fro") / np.linalg.norm(A, "fro")
    print(f"k={k}: storage={k * (m + n + 1)}, relative error={error:.4f}")
```

The printed storage count is $k(m+n+1)$. Exact error and visual quality depend on the chosen image. As k increases, more detail is restored and the approximation error cannot increase, but more values must be stored.